

Lumma Stealer Network Traffic Analysis Report

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Table of Contents

1. Executive Summary
2. Objectives
3. Environment
4. Tools Used
5. Attack Flow
6. Network Traffic Analysis
 - Protocol Hierarchy
 - IPv4 Endpoints
 - IPv4 Conversations
 - DNS Analysis
 - TLS Handshake
 - Command-and-Control Communication
 - Follow TCP Stream
 - I/O Graph
 - Protocol Hierarchy (TLS)
 - TCP Sequence Graph
7. Exported Objects
8. Expert Information
9. Indicators of Compromise
10. MITRE ATT&CK Mapping

- 11. Detection Opportunities
- 12. Infection Timeline
- 13. Conclusion
- 14. References

1. Executive Summary

This report documents the forensic analysis of a Lumma Stealer malware infection using Wireshark. The objective was to reconstruct the malware's network behavior, identify malicious infrastructure, recover indicators of compromise (IoCs), and document the attack chain observed in the provided packet capture.

2. Objectives

- Identify malicious DNS activity.
- Detect Command-and-Control communication.
- Analyze encrypted TLS traffic.
- Recover downloaded malware payloads.
- Extract Indicators of Compromise.
- Map observations to MITRE ATT&CK.

3. Environment

- Operating System: Windows
- Analysis Tool: Wireshark
- Sample: Lumma Stealer PCAP
- Analyst: Agata Gabara

4. Tools Used

- Wireshark
- VirusTotal
- Any.Run
- Hybrid Analysis

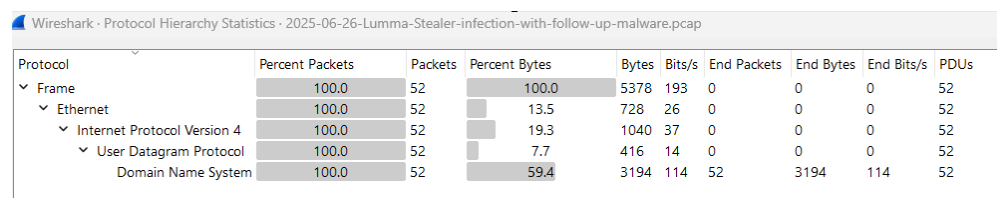
5. Attack Flow

(Insert attack flow diagram)

The infection begins with DNS resolution of the malicious domain, followed by encrypted TLS communication with the Command-and-Control server. During the session, the malware retrieves an additional executable payload over HTTP.

6. Network Traffic Analysis

Protocol Hierarchy

A screenshot of the Wireshark Protocol Hierarchy Statistics window. The title bar reads "Wireshark - Protocol Hierarchy Statistics - 2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap". The table shows the following data:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
▼ Frame	100.0	52	100.0	5378	193	0	0	0	52
▼ Ethernet	100.0	52	13.5	728	26	0	0	0	52
▼ Internet Protocol Version 4	100.0	52	19.3	1040	37	0	0	0	52
▼ User Datagram Protocol	100.0	52	7.7	416	14	0	0	0	52
Domain Name System	100.0	52	59.4	3194	114	52	3194	114	52

The Protocol Hierarchy view confirms that the filtered capture contains exclusively DNS traffic encapsulated within UDP over IPv4. DNS accounts for all 52 packets and approximately 59.4% of the transmitted bytes, indicating that the applied filter successfully isolates the domain resolution phase of the malware communication.

IPv4 Endpoints

2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

Wireshark · Endpoints · 2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap

Endpoint Settings

- ☐ Name resolution
- ☐ Display raw data
- ☐ Hide aggregated
- ☒ Limit to display filter

Copy

Map

Protocol

- ☐ Bluetooth
- ☐ BPv7
- ☐ DCCP
- ☐ DNP 3.0
- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☐ IEEE 802.15.4
- ☐ ILNP
- ☒ IPv4
- ☒ IPv6
- ☐ IPX
- ☐ JXTA

Ethernet · 2		IPv4 · 2		IPv6	TCP	UDP · 7								
Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
10.6.26.1	52	5 kB	52	100.00%	26	3 kB	26	2 kB						
10.6.26.101	52	5 kB	35,211	0.15%	26	2 kB	26	3 kB						

1509 × 857px Size: 59.1KB

The IPv4 Endpoints view confirms that the DNS traffic is limited to communication between the infected host (10.6.26.101) and the local DNS resolver (10.6.26.1). The workstation transmitted 26 DNS queries and received 26 responses, resulting in 52 packets exchanged. This represents the domain resolution phase that precedes communication with the malware's external infrastructure.

IPv4 Conversations

Wireshark · Conversations · 2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap

Conversation Settings

- ☐ Absolute start time
- ☐ Display raw data
- ☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

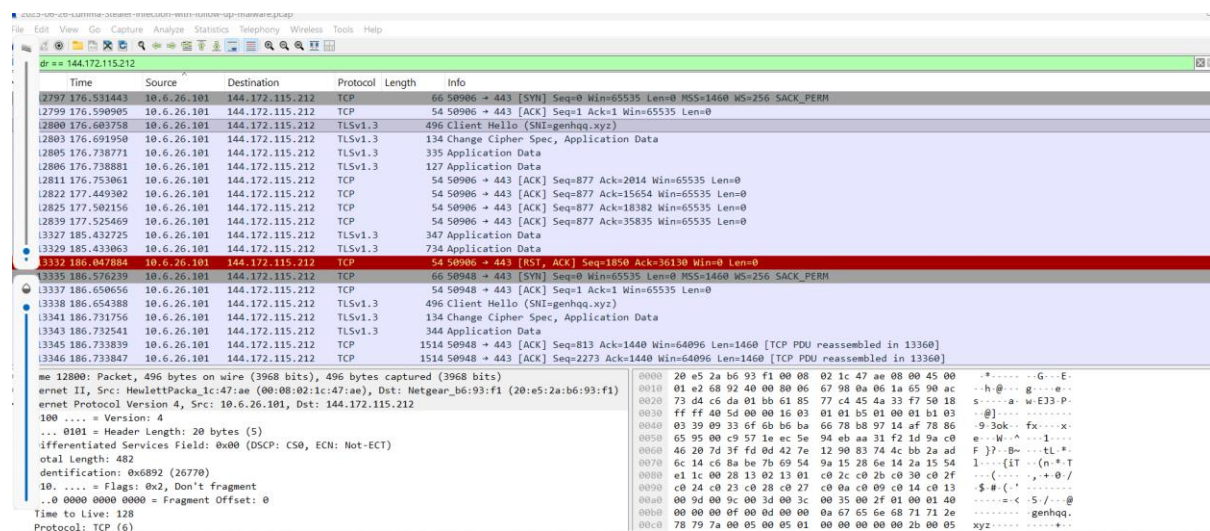
Protocol

- ☐ Bluetooth
- ☐ BPv7
- ☐ DCCP
- ☐ DNP 3.0
- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☐ IEEE 802.15.4
- ☐ ILNP
- ☒ IPv4
- ☒ IPv6
- ☐ IPX
- ☐ JXTA

Ethernet · 1		IPv4 · 1		IPv6	TCP	UDP · 6								
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.6.26.101	10.6.26.1	52	5 kB	0	52	100.00%	26	2 kB	26	3 kB	0.000000	222.6502	71 bits/s	121 bits/s

The IPv4 Conversations statistics show a single DNS conversation between the infected workstation (10.6.26.101) and the local DNS resolver (10.6.26.1). The session consists of 52 packets exchanged over approximately 223 seconds, with an equal number of DNS queries and responses. This communication represents the initial stage of the infection, where the malware resolves external domains before establishing encrypted Command-and-Control (C2) connections.

Command-and-Control Communication



Time	Source	Destination	Protocol	Length	Info
12797.176.531443	10.6.26.101	144.172.115.212	TCP	66	50906 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
12799.176.590905	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
12800.176.603758	10.6.26.101	144.172.115.212	TLSv1.3	496	Client Hello (SNI=genhq.xyz)
12803.176.691950	10.6.26.101	144.172.115.212	TLSv1.3	134	Change Cipher Spec, Application Data
12805.176.738771	10.6.26.101	144.172.115.212	TLSv1.3	335	Application Data
12806.176.738881	10.6.26.101	144.172.115.212	TLSv1.3	127	Application Data
12811.176.753061	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [ACK] Seq=877 Ack=2014 Win=65535 Len=0
12822.177.449302	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [ACK] Seq=877 Ack=15654 Win=65535 Len=0
12825.177.502156	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [ACK] Seq=877 Ack=18382 Win=65535 Len=0
12839.177.525469	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [ACK] Seq=877 Ack=35835 Win=65535 Len=0
13327.185.432725	10.6.26.101	144.172.115.212	TLSv1.3	347	Application Data
13329.185.433063	10.6.26.101	144.172.115.212	TLSv1.3	734	Application Data
13332.186.441455	10.6.26.101	144.172.115.212	TCP	54	50906 → 443 [RST] Seq=1850 Ack=36130 Win=0 Len=0
13335.186.576239	10.6.26.101	144.172.115.212	TCP	66	50948 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
13337.186.650656	10.6.26.101	144.172.115.212	TCP	54	50948 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
13338.186.654388	10.6.26.101	144.172.115.212	TLSv1.3	496	Client Hello (SNI=genhq.xyz)
13341.186.731756	10.6.26.101	144.172.115.212	TLSv1.3	134	Change Cipher Spec, Application Data
13343.186.732541	10.6.26.101	144.172.115.212	TLSv1.3	344	Application Data
13345.186.733839	10.6.26.101	144.172.115.212	TCP	1514	50948 → 443 [ACK] Seq=813 Ack=1440 Win=64096 Len=1460 [TCP PDU reassembled in 13360]
13346.186.733847	10.6.26.101	144.172.115.212	TCP	1514	50948 → 443 [ACK] Seq=2273 Ack=1440 Win=64096 Len=1460 [TCP PDU reassembled in 13360]

12800: Packet, 496 bytes on wire (3968 bits), 496 bytes captured (3968 bits) on interface 0
 Ethernet II, Src: Hewlett-Packard (08:00:02:1c:47:ae), Dst: Netgear (08:00:0c:20:70:00) (20:e5:2a:b6:93:f1)
 Internet Protocol Version 4, Src: 10.6.26.101, Dst: 144.172.115.212
 100 ... = Version: 4
 ... 0101 = Header Length: 20 bytes (5)
 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 482
 Identification: 0x6092 (26770)
 10 ... = Flags: 0x2, Don't fragment
 ... 0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 128
 Protocol: TCP (6)

This filtered view captures the malware's communication with its Command-and-Control (C2) infrastructure. The infected workstation (**10.6.26.101**) establishes a TCP connection to **144.172.115.212** on port **443**, followed by a TLS Client Hello exposing the Server Name Indication (SNI) **genhq.xyz**. The subsequent TLS Application Data packets confirm encrypted data exchange with the remote server. After the session is reset (RST), the malware immediately initiates a new TLS connection, demonstrating persistent communication attempts with the C2 infrastructure.

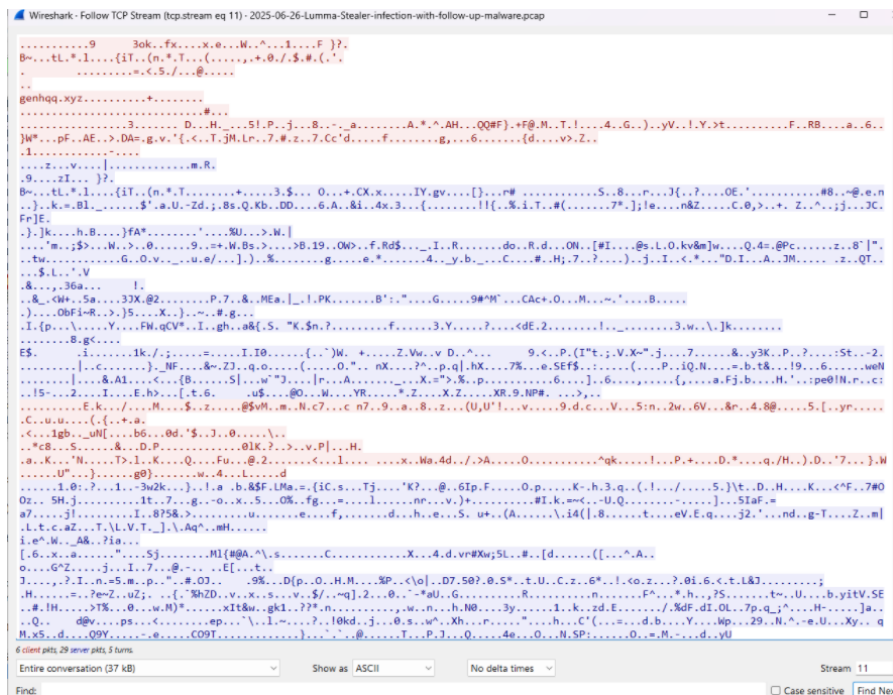
TLS Handshake

2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap					
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
[tls.handshake.extensions.server_name == 'genhq.xyz']					
No.	Time	Source	Destination	Protocol	Length Info
12800	176.603758	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
13338	186.654388	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
14555	191.852277	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
14573	193.197646	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
16175	198.493542	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
16287	199.873511	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
16383	201.416946	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)
18689	206.398827	10.6.26.101	144.172.115.212	TLSv1.3	496 Client Hello (SNI=genhq.xyz)

> Frame 12800: Packet, 496 bytes on wire (3968 bits), 496 bytes captured (3968 bits) on interface 0 > Ethernet II, Src: Hewlett-Packard_10:00:00:08:00:00, Dst: Netgear_08:00:27:00:00:00 (20:e5:2a:b6:93:f1) > Internet Protocol Version 4, Src: 10.6.26.101, Dst: 144.172.115.212 0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total length: 482 Identification: 0x6892 (26770) > 010. = Flags: 0x2, Don't fragment ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 128 Protocol: TCP (6) Header Checksum: 0x6798 (validation disabled) [Header checksum status: Unverified] Source Address: 10.6.26.101 Destination Address: 144.172.115.212 [Stream index: 10] > Transmission Control Protocol, Src Port: 50906, Dst Port: 443, Seq: 1, Ack: 1, Len: 442 > Transport Layer Security		0000 20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00G...E: 0010 01 e2 68 92 40 00 00 06 67 98 0a 06 1a 65 90 ac ..h@...g...e- 0020 73 d4 c6 da 01 bb 61 85 77 c4 45 4a 33 f7 50 18 s...a-w-E3-P- 0030 ff ff 40 5d 00 00 16 03 01 01 b5 01 00 01 b1 03 --[]...-...- 0040 03 39 09 33 6f 6b b6 ba 66 78 b8 97 14 af 78 86 .9:3ok..fx...x- 0050 65 95 00 c9 57 1e ec 5e 94 eb aa 31 f2 1d 9a c0 e...W-^...1...- 0060 46 20 7d 3f fd 0d 42 7e 12 90 83 74 4c bb 2a ad F }?-B-...tL*. 0070 6c 14 c6 8a be 7b 69 54 9a 15 28 6e 14 2a 15 54 1---{1T---(n..T 0080 e1 1c 00 28 13 02 13 01 c0 2c c0 2b c0 30 c0 2f ---(-...+*0/ 0090 c0 24 c0 23 c0 28 c0 27 c0 0a c0 09 c0 14 c0 13 \$.#(-...- 00a0 00 9d 00 9c 00 3d 00 3c 00 35 00 2f 01 00 01 40<S/-...@ 00b0 00 00 00 0f 00 0d 00 00 0a 67 65 6e 68 71 71 2egenhq. 00c0 78 79 7a 00 05 00 05 01 00 00 00 00 00 2b 00 05 xyz.....+...+ 00d0 04 03 04 03 03 00 0d 00 1a 00 18 00 04 00 05 00-...- 00e0 06 04 01 05 01 02 01 04 03 05 03 02 03 02 06-...- 00f0 01 06 03 00 23 00 00 00 0a 00 08 00 06 00 1d 00#...-...- 0100 17 00 18 00 0b 00 02 01 00 00 33 00 00 ce 00-3...-...- 0110 1d 00 20 44 81 8f bc 48 1e 5f ba fc e4 35 21 1d ..D--H...-51- 0120 50 14 b0 6a d9 dd b0 38 ce 1b 2d cd 5f 61 93 18 P-j--B...-a- 0130 c5 f1 13 00 17 00 41 04 2a 01 5e 08 41 48 85 d6A:*^AH.. 0140 c9 51 51 23 46 7d ee 2b 46 40 a2 4d 00 ea 54 17 QQ#F)+F@-M--T- 0150 21 fc 83 95 ba 34 94 f0 47 b0 91 29 8a f7 79 56 [---4---G---yV 0160 04 f5 21 ef 59 91 3e 74 b6 a8 04 9d a5 b3 a6 ed ...Y>T.....
---	--	--

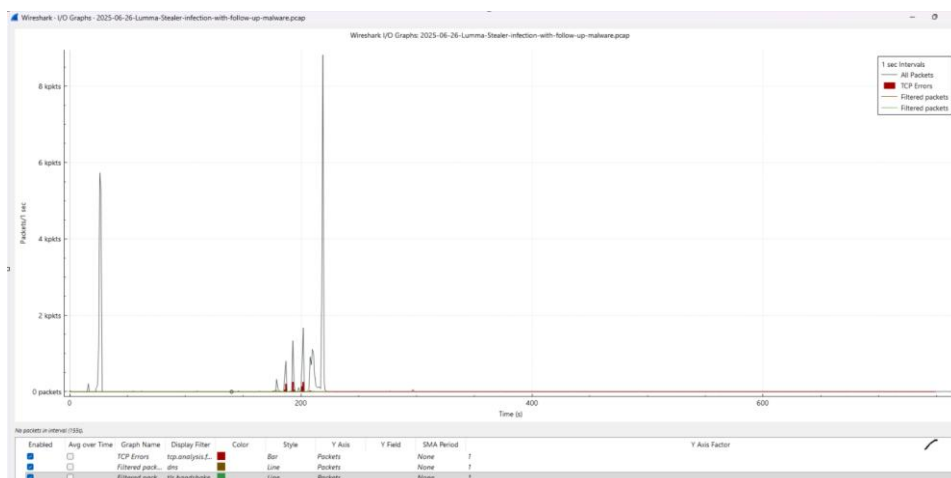
This view isolates all TLS Client Hello packets containing the Server Name Indication (SNI) **genhq.xyz**. The repeated appearance of the same hostname throughout the capture indicates that the malware persistently attempted to establish encrypted connections with its Command-and-Control (C2) infrastructure. While the TLS payload remains encrypted, the SNI extension exposes the destination domain, making it a valuable Indicator of Compromise (IoC) for network detection and threat hunting.

Follow TCP Stream



The reconstructed TCP stream contains the complete bidirectional communication exchanged during the TLS session. The visible plaintext hostname `genhq.xyz` originates from the TLS Client Hello message, where the Server Name Indication (SNI) extension specifies the intended destination. Following the handshake, all subsequent application data is encrypted, preventing direct analysis of the malware payload while confirming the establishment of a secure communication channel with the C2 server.

I/O Graph



The I/O Graph illustrates packet rates over time during the captured malware infection. Short bursts of network activity are visible during the initial DNS lookup, TLS handshake, and

Protocol Hierarchy (TLS)

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
▼ Frame	100.0	49	100.0	40640	34 k	0	0	0	49
▼ Ethernet	100.0	49	1.7	686	576	0	0	0	49
▼ Internet Protocol Version 4	100.0	49	2.4	980	823	0	0	0	49
▼ Transmission Control Protocol	100.0	49	2.5	996	837	36	736	618	49
Transport Layer Security	26.5	13	93.4	37978	31 k	13	37978	31 k	13

TLS Handshake Details

```

Wireshark 2.10.0 - 60 Luma Data Infection with follow up malware.pcap
File Edit View Go Capture Analysis Statistics Help
[Shandhu]

No.    Time Source                Destination              Protocol Length Info
-----
12692 62.886952   10.6.26.101             157.240.24.35           TLSv1.2 258 Client Hello (SHA+www.facebook.com)
12693 62.910379   157.240.24.35           10.6.26.101            TLSv1.2 195 Server Hello, Change Cipher Spec, Encrypted Handshake Message
12695 62.915719   10.6.26.101            157.240.24.35           TLSv1.2 105 Change Cipher Spec, Encrypted Handshake Message
12710 18.924479   10.6.26.101            52.123.128.14          TLSv1.3 195 Client Hello (SHA+ecs.office.com)
12712 18.962343   52.123.128.14          10.6.26.101            TLSv1.3 1430 Server Hello, Change Cipher Spec
12718 18.680751   10.6.26.101            52.123.128.14          TLSv1.2 225 Client Hello (SHA+windows.msn.com)
12780 164.726239 10.6.26.101            144.172.51.83          TLSv1.2 1514 Server Hello
12781 164.726340 10.6.26.101            144.172.51.83          TLSv1.2 1451 Certificate, Server Key Exchange, Server Hello Done
12783 164.729748 10.6.26.101            144.172.51.83          TLSv1.2 239 Client Hello (SHA+clientservices.googleapis.com)
12785 164.769692 10.6.26.101            144.172.51.83          TLSv1.2 312 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
12800 176.603758 10.6.26.101            144.172.51.212         TLSv1.3 490 Client Hello (SHA+genhq.xyz)
12802 176.675047 144.172.51.212         10.6.26.101            TLSv1.3 1493 Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, Application Data
12803 176.675297 10.6.26.101            144.172.51.212         TLSv1.2 239 Client Hello (SHA+clientservices.googleapis.com)
12859 178.998372 10.6.26.101            142.250.113.84         TLSv1.2 229 Client Hello (SHA+accounts.google.com)
12860 179.020721 142.250.113.84        10.6.26.101            TLSv1.2 1514 Server Hello
12863 179.020722 142.250.113.84        10.6.26.101            TLSv1.2 235 Certificate, Server Key Exchange, Server Hello Done
12865 179.020747 10.6.26.101            142.250.113.84         TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
12870 179.041236 142.250.113.84        10.6.26.101            TLSv1.2 1514 Server Hello
12871 179.041237 142.250.113.84        10.6.26.101            TLSv1.2 1334 Certificate, Server Key Exchange, Server Hello Done
12873 179.055575 142.250.113.84        TLSv1.2 147 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message

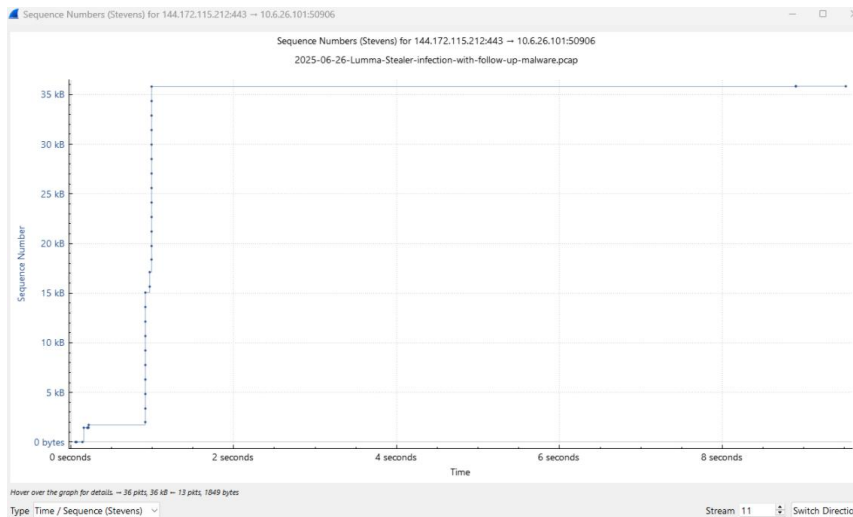
# Frame 12785: Packet, 312 bytes on wire (2406 bits), 312 bytes captured (2406 bits)
# Ethernet II, Src: Netgear_b6:93:f1 (20:es:ba:b6:93:f1), Dst: HualtekPa...i:47:aa (08:00:02:1c:47:aa)
# Internet Protocol Version 4, Src: 23.47.51.83, Dst: 10.6.26.101
0100 ... = Version: 4
... .. Offset = Header Length: 20 bytes (5)
# Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 298
Identification: 0x5655 (21865)
... .. Flags: b0f
... .. 0.0000 0000 0000 = Fragment Offset: 0
... .. To Live: 118
Protocol: TCP (6)
Header Checksum: 0x75fc [validation disabled]
Header checksum status: Unverified
Source Address: 23.47.51.83
Destination Address: 10.6.26.101
[Stream index: 9]
# Transmission Control Protocol, Src Port: 443, Dst Port: 50903, Seq: 2858, Ack: 298, Len: 258
# Transport Layer Security

```

8

the malicious domain genhqq.xyz. The Server Name Indication (SNI) field exposes the destination hostname before encryption is established, making it possible to identify the Command-and-Control (C2) infrastructure despite the encrypted communication.

TCP Sequence Graph



The initial spike in sequence numbers is consistent with the exchange of encrypted TLS application data between the infected host and the C2 server. No packet retransmissions or abnormal TCP behavior are visible, indicating a stable and successful communication session.

The TCP Time-Sequence (Stevens) graph illustrates the progression of sequence numbers during the communication between the infected workstation (10.6.26.101) and the Command-and-Control (C2) server (144.172.115.212). The rapid increase in sequence numbers during the first second indicates a burst of encrypted TLS data transmission. After the initial exchange, the sequence numbers stabilize, suggesting that the malware completed the primary data transfer and maintained the TCP session with minimal additional traffic.

7. Exported Objects

HTTP Objects

Packet	Hostname	Content Type	Size	Filename
12867	clients2.google.com	application/json	81 bytes	current?cup2key=9:PFGa0hZkXKY2b3DsWKtGmyVZ
23387	86.54.25.50	application/octet-stream	5943 kB	soks.exe

The HTTP Object List reveals two downloadable objects captured during the network session. The most significant artifact is the executable file **soks.exe** (approximately **5.9 MB**) downloaded from the host **86.54.25.50** with the content type **application/octet-stream**. This confirms the malware delivery stage of the attack and demonstrates that the payload can be recovered directly from the packet capture for further static or dynamic malware analysis.

Recovered executable:

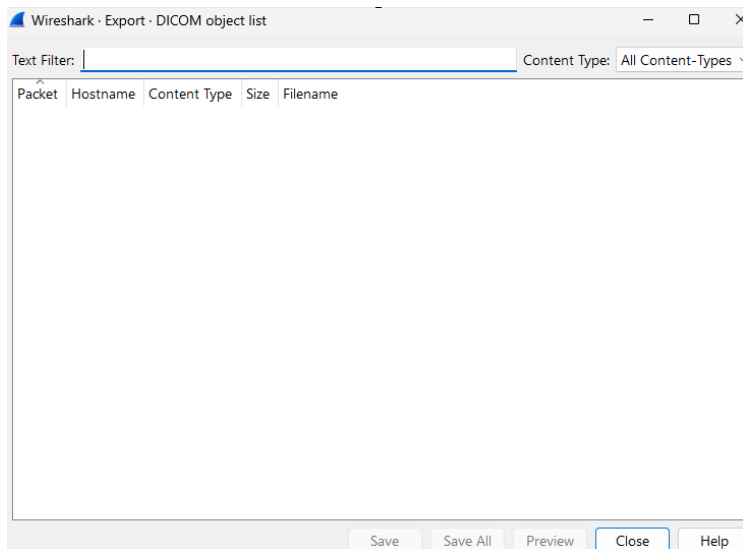
- soks.exe

SMB Objects

Packet	Hostname	Content Type	Size	Filename
--------	----------	--------------	------	----------

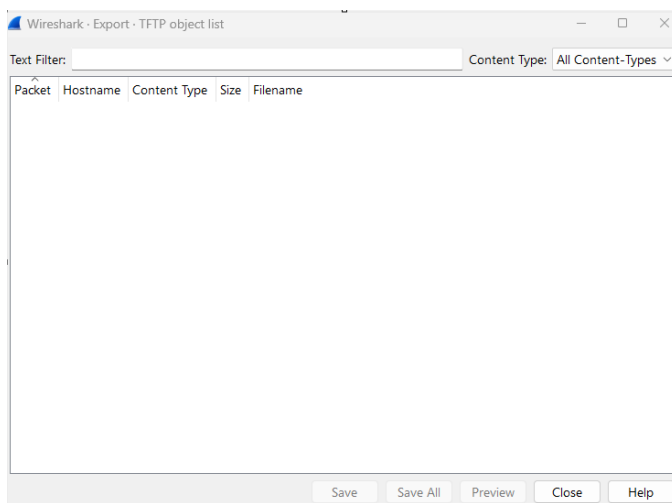
No exported SMB objects detected.

DICOM Objects



No exported DICOM objects detected.

TFTP Objects



No exported TFTP objects detected.

8. Expert Information

Wireshark - Expert Information - 2025-06-26-Lumma-Stealer-infection-with-follow-up-malware.pcap				
Severity	Summary	Group	Protocol	Count
> Warning	Connection reset (RST)	Sequence	TCP	1
> Note	The SYN packet does not contain a SACK PERM option	Protocol	TCP	1
> Chat	This legacy_version field MUST be ignored. The supported_versions extensi...	Deprecated	TLS	2
> Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP	1
> Chat	Connection establish request (SYN)	Sequence	TCP	1

Minor TCP warnings were observed during the communication. No critical protocol errors affected the investigation.

9. Indicators of Compromise

Typ	Wartość
Domena	genhqq.xyz
IP	144.172.115.212
Plik	soks.exe
Host pobrania	86.54.25.50
Port	443/TCP
Malware	Lumma Stealer

10. MITRE ATT&CK Mapping

Tactic Technique

Command and Control T1071.001

Command and Control T1573

Command and Control T1105

11. Detection Opportunities

- Monitor DNS queries for malicious domains.
- Inspect TLS Server Name Indication (SNI).
- Detect outbound HTTPS connections to rare destinations.
- Monitor executable downloads over HTTP.

12. Infection Timeline

1. DNS query for genhqq.xyz
2. TCP three-way handshake
3. TLS Client Hello
4. Encrypted communication established
5. Malware payload retrieved
6. Persistent C2 communication

13. Conclusion

The investigation confirmed communication with the Lumma Stealer Command-and-Control infrastructure using encrypted TLS sessions. Network metadata, DNS activity, exported HTTP objects, and protocol statistics were sufficient to reconstruct the attack timeline and identify the malware infrastructure despite encrypted application traffic.

14. References

- Wireshark Documentation
- MITRE ATT&CK
- Malware-Traffic-Analysis.net
- VirusTotal